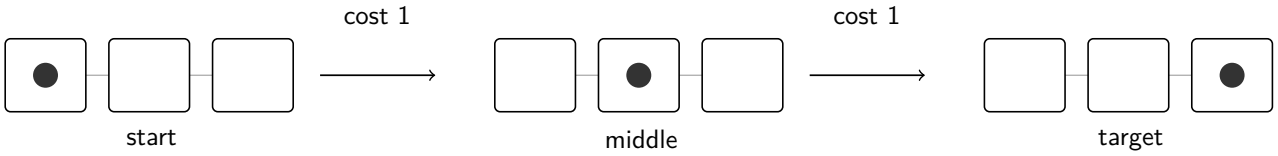
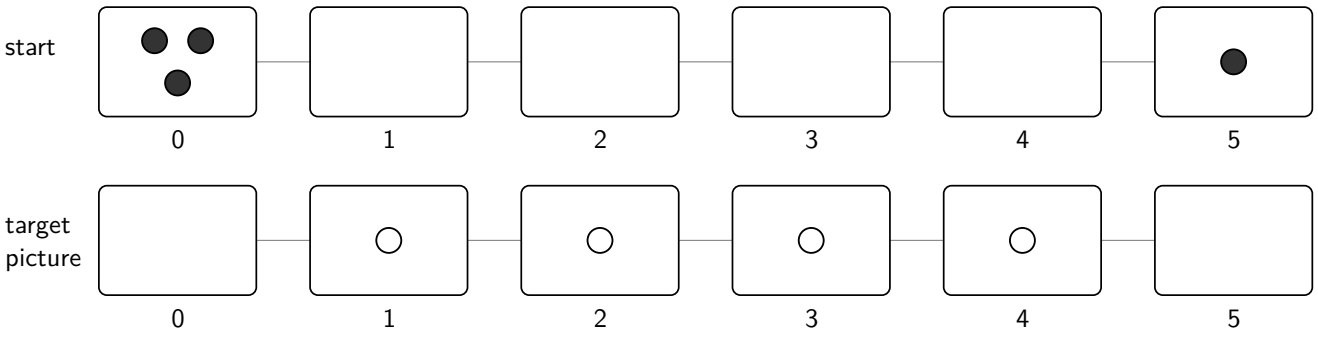
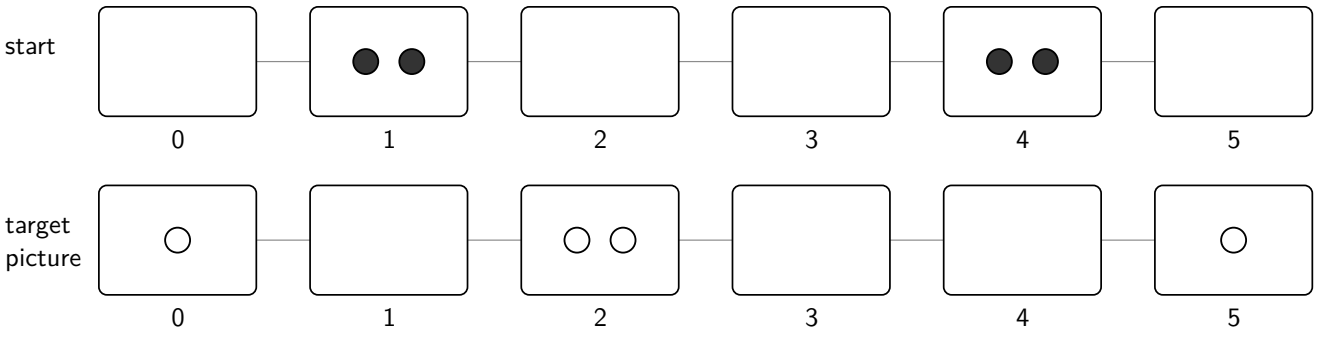
Move one counter at a time to a neighboring bin. Each crossing costs 1. Bins hold any number of counters, and counters may pass through occupied bins. Any counter may fill any target place. Only start counters move; the target picture stays fixed. Keep one record of your moves at a time so you can count the work afterward.

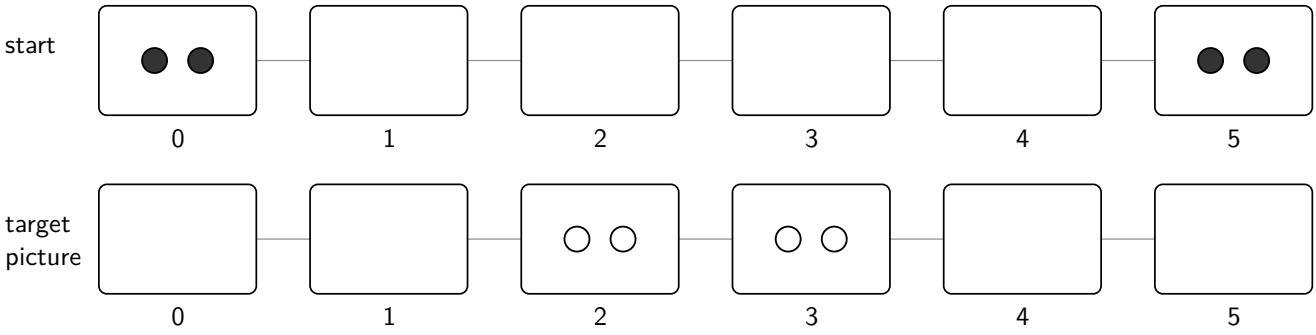


Problem 1: Make each start match its target using the least work you can. Find two best move records for each pair. Changing the order of moves counts as different.

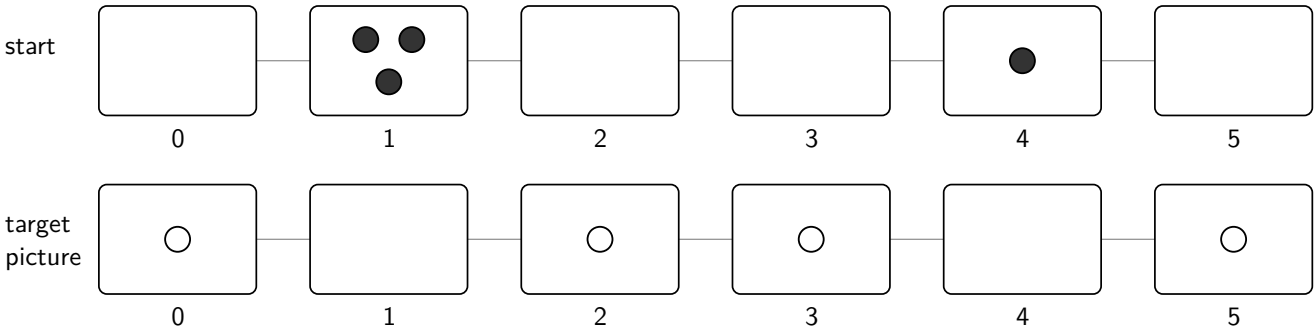




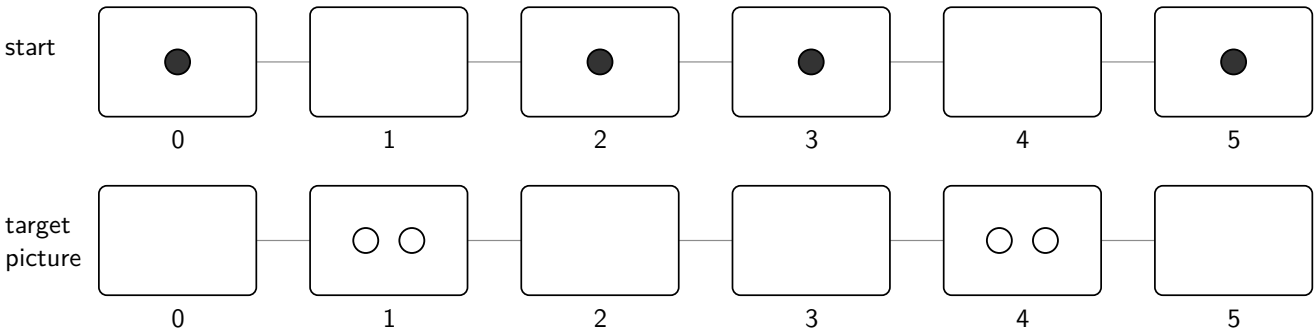
Problem 2: Find the least work for each pair. Explain why no plan can use less.



work: _____

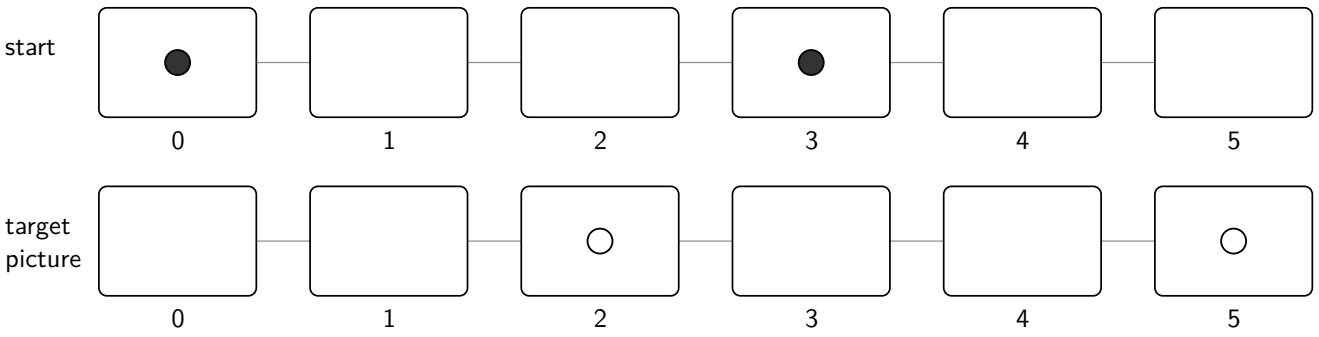


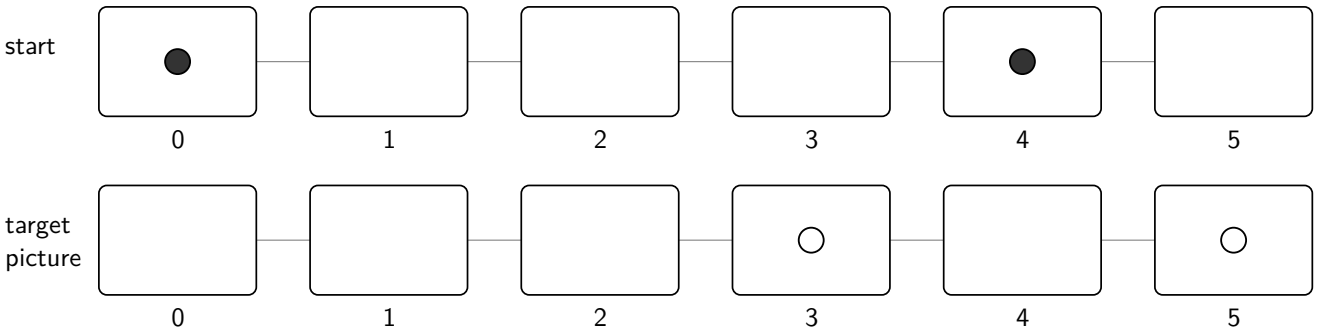
work: _____



work: _____

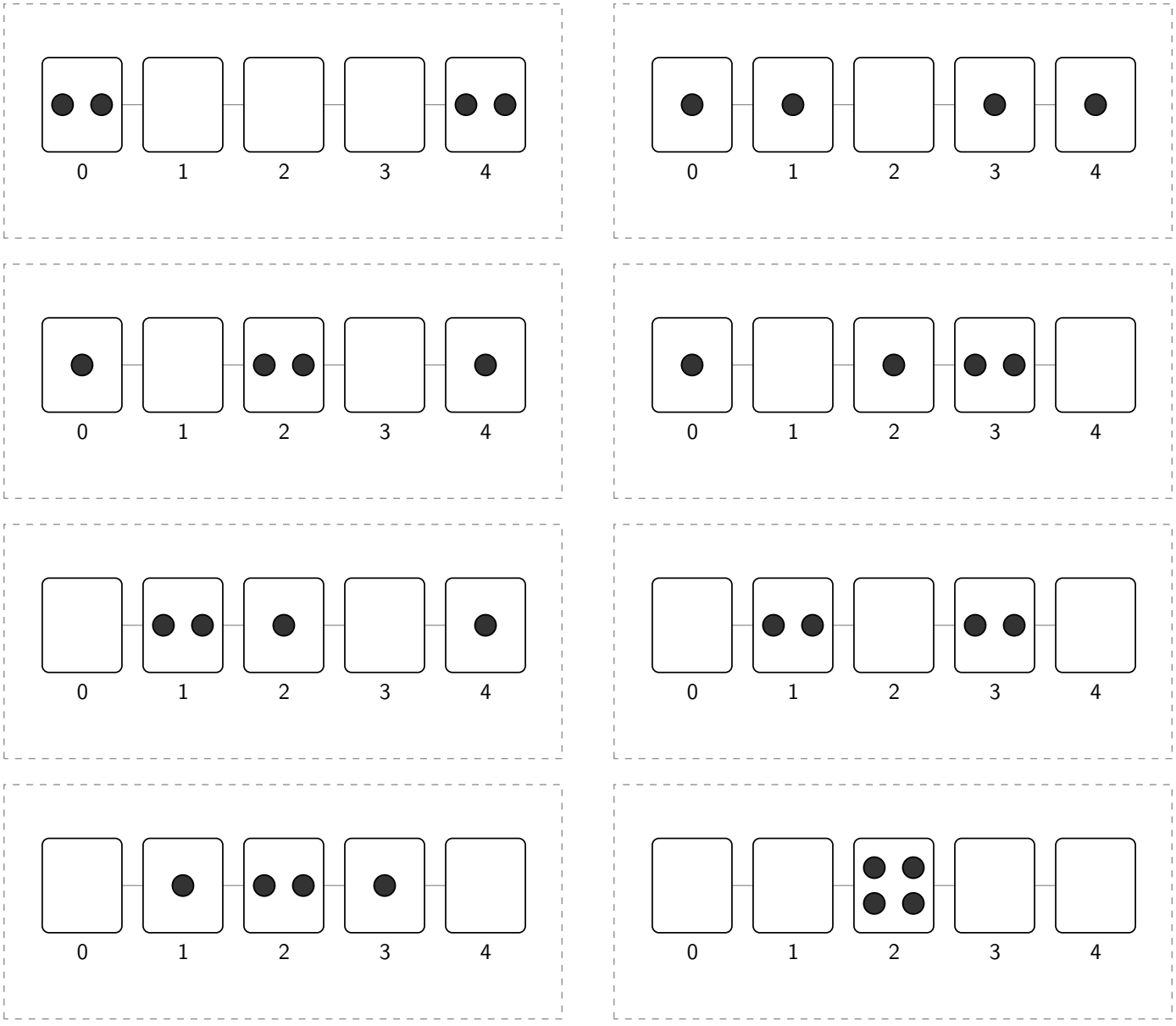
Problem 3: A delivery takes one counter that has not yet been delivered all the way to one target place that has not yet been filled. Once delivered, that counter stays there. A rule says to make the shortest remaining delivery each time. Test this rule on both pairs. Does it always give the least work? Can tied choices change the result?





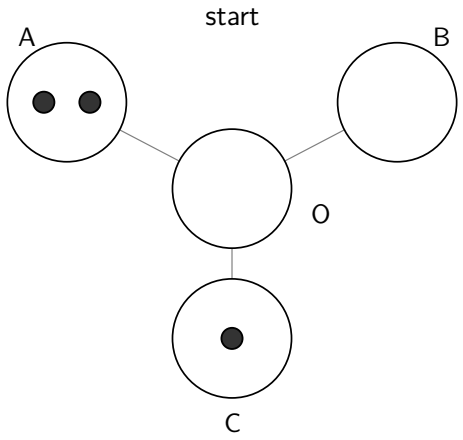
One counter in bin 1 and one in bin 3 have position total $1 + 3 = 4$.

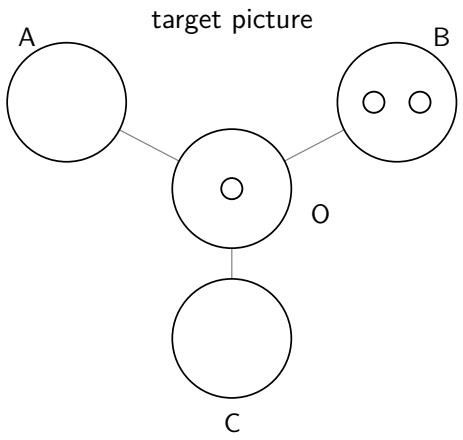
Problem 4: Each card has four counters with position total 8. Which two need the most work to change one into the other? Explain why no pair needs more. Is any four-counter arrangement on bins 0 to 4 with position total 8 missing?



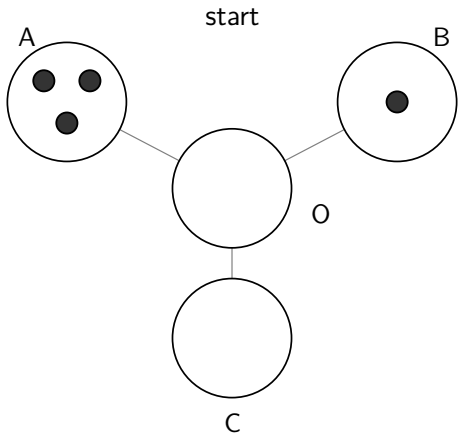
work:

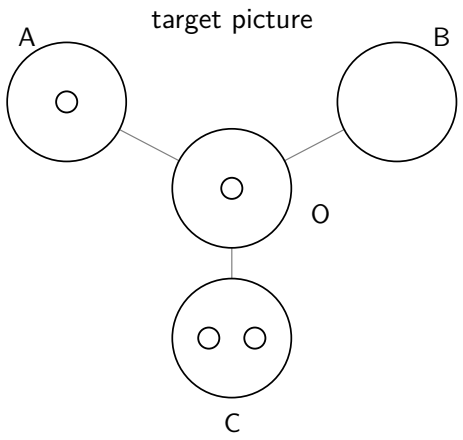
Problem 5: Move counters along the roads. One counter crossing one road costs 1. Find the least work for both pairs. Find a way to know the least work before moving. Does it work for every start and target with the same number of counters on these roads?





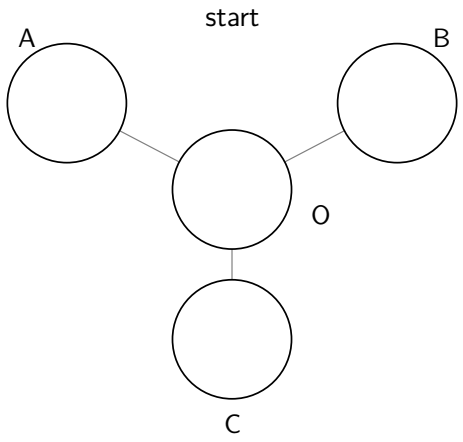
work: _____

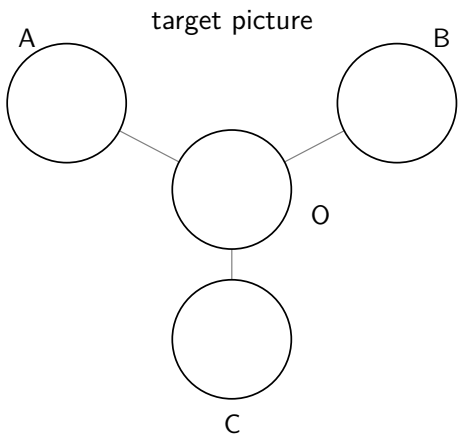




work: _____

Make a start and target with four counters each to test your way.





Problem 6: Make three arrangements, A, B, and C, with four counters in each. Can the least work from A to B plus the least work from B to C be less than the least work straight from A to C? Explain. Find examples where the amounts are equal and where they are different.

